SEMINAR p-ADIC HODGE THEORY

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1. General introduction and motivation

The absolute Galois group of the rational numbers $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is a central object of study in number theory. This profinite group is still quite mysterious. A key idea in modern number theory is to study this group via its representations. The study of representations of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ can then be approached via restricting representations to the subgroups $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$[1] for varying primes p. In this seminar, we will consider continuous representations of Galois groups such as $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ on finite-dimensional $\mathbb{Q}_\ell$-vector space. We will see that for primes $\ell \neq p$, these representations are quite a bit simpler than in the case $\ell = p$ (see §4). It is the latter case, when $\ell = p$, that is the main concern of p-adic Hodge theory.

p-adic Hodge theory was initiated by John Tate in the sixties, with early ideas due to Alexandre Grothendieck. The most substantial developments were due to Jean-Marc Fontaine, and the theory was developed further by many mathematicians. It has now become an indispensable tool in modern number theory; it was, for example, an important tool in the proof of Fermat's Last Theorem by Andrew Wiles, with a crucial step by Richard Taylor and Wiles, in the nineties.

The name of this subject can be explained as follows: let X be a smooth projective algebraic variety over the field of rational numbers $\mathbb{Q}$ (in other words: we look at the solution set of a family of homogeneous polynomials whose coefficients are rational numbers and whose Jacobian matrix very roughly speaking has maximal possible rank). Let $X(\mathbb{C})$ be the solution set over the complex numbers. This admits the structure of a compact complex manifold. We may associate the following two $\mathbb{Q}$-vector spaces to this situation: the singular cohomology $H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Q})$ and the de Rham cohomology $H^n_{\mathrm{dR}}(X/\mathbb{Q})$. The de Rham comparison isomorphism says that

$$H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} \simeq H^n_{\mathrm{dR}}(X/\mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}. \tag{1}$$

This isomorphism is constructed from the perfect pairing between singular *homology* and de Rham *cohomology*

$$H_n(X(\mathbb{C}), \mathbb{C}) \times H^n_{\mathrm{dR}}(X_{\mathbb{C}}/\mathbb{C}) \to \mathbb{C}$$

given by integrating differential forms along paths

$$(\gamma, \omega) \mapsto \int_\gamma \omega$$

and using the duality $H_n(X(\mathbb{C}), \mathbb{C})^\vee \simeq H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$. Let $\{c_1, ..., c_m\}$ and $\{\omega_1, ...\omega_m\}$ be $\mathbb{Q}$-bases of the vector space $H^n_{\mathrm{sing}}(X(\mathbb{C}), \mathbb{Q})$ and $H^n_{\mathrm{dR}}(X/\mathbb{Q})$ respectively. We obtain a dual basis $\{\gamma_1, ..., \gamma_m\}$ of $H_n(X(\mathbb{C}), \mathbb{Q})$. Then we may describe the isomorphism in (1) by a matrix $(b_{i,j})_{1 \leq i,j \leq m} \in \mathrm{GL}_m(\mathbb{C})$. The entries $b_{i,j} \in \mathbb{C}$ are called the periods of X; they depend of course on the choice of the basis since they are given by

$$b_{i,j} = \int_{\gamma_i} \omega_j,$$

but for example the field $\mathbb{Q}(b_{i,j} : 1 \leq i, j \leq m) \subset \mathbb{C}$ generated by the periods just depends on X. Periods are therefore values of integrals of algebraic differential forms along paths and some of the examples (if we stop insisting on *projective* varieties) that show up are: $2\pi i = \int_{|z|=1} \frac{dz}{z}$ (corresponding to the affine line minus the origin) and $\frac{\Gamma(1/4)\Gamma(1/2)}{\Gamma(3/4)} = 2\int_1^{+\infty} \frac{dx}{\sqrt{x^3 - x}}$ (elliptic curve $y^2 = x^3 - x$). We therefore see that the field of complex numbers $\mathbb{C}$ contains *all* periods of algebraic varieties.

Date: September 18, 2026.

[1]Strictly speaking, this embedding depends on the choice of an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$ and is well-defined only up to conjugation.

The geometric aspect of p-adic Hodge theory is the quest to find out to what extent a similar story also exists if we replace smooth projective varieties over the field $\mathbb{Q}$ by smooth projective varieties over the field $\mathbb{Q}_p$.

The varieties we will mostly consider in this seminar are elliptic curves. These are smooth projective 1-dimensional algebraic varieties given over fields K with $\mathrm{char}(K) \neq 2, 3$ in affine coordinates by one equation of the form $E : y^2 = x^3 + ax + b$ with $a, b \in K$ and $\Delta = -16(4a^3 + 27b^2) \neq 0$. These curves have the beautiful feature that one may add and subtract points in the solution set $E(L)$, L/K a field extension, making it into an abelian group. In §6 we will learn a bit more about them. One of the leading examples of a Galois representation in our seminar is that of the *ℓ-adic Tate module* of an elliptic curve. It is the ℓ-adic analogue of $H_1(E(\mathbb{C}), \mathbb{Z})$.

To explain what this is, let us assume that we are considering an elliptic curve with $a, b \in \mathbb{Q}_p$, let ℓ be a prime, then the ℓ-adic Tate module is given by $T_\ell(E) = \lim_\ell E(\overline{\mathbb{Q}}_p)[\ell^n]$. This is a finite free $\mathbb{Z}_\ell$-module of rank two which admits a continuous action by $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. Algebraic differential forms still make sense over $\mathbb{Q}_p$. In the case of an elliptic curve E over $\mathbb{Q}_p$ given by an equation $E : y^2 = x^3 + ax + b$ as above, $H^1_{\mathrm{dR}}(E/\mathbb{Q}_p)$ is a 2-dimensional $\mathbb{Q}_p$-vector space with one possible basis given by $\omega_1 = \frac{dx}{y}$ and $\omega_2 = x\frac{dx}{y}$. The analogue of the field of complex numbers in the p-adic world is $\mathbb{C}_p = \widehat{\overline{\mathbb{Q}}}$; we will study it in §5. It has a natural continuous action by the absolute Galois group $G_{\mathbb{Q}_p} = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. The quest for an analogue of the pairing underlying the isomorphism (1) would be to try to find a perfect $G_{\mathbb{Q}_p}$-*equivariant* pairing

$$(T_p(E) \otimes_{\mathbb{Z}_p} \mathbb{C}_p) \times H^1_{\mathrm{dR}}(E/\mathbb{C}_p) \to \mathbb{C}_p.$$

In the first talk 2, I will explain how this would lead to the existence of an element $(2\pi i)_p \in \mathbb{C}_p$ with a very special property with respect to the action by the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. In §5, we will see that such an element cannot exist. This implies that the field of complex p-adic numbers does *not* contain enough periods to make an isomorphism as in (1) hold.

Much of the hard work that will be done in this seminar (see §8, §9, §10, §11, §12) is concerned with the ingenious constructions due to Fontaine of certain topological $\mathbb{Q}_p$-algebras B (rings of p-adic periods) supporting a continuous $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$-action. These rings B contain enough elements to make p-adic analogues of the de Rham comparison in (1) true. For example, for one such $B = \mathbb{B}_{\mathrm{dR}}$ and an elliptic curve $E/\mathbb{Q}_p$, there is an isomorphism

$$(V_p(E))^\vee \otimes_{\mathbb{Q}_p} B \simeq H^1_{\mathrm{dR}}(E/\mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B,$$

where $V_p(E) = T_p(E) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ and which is compatible with the Galois action and certain natural filtrations. These rings B will also allow us to group the collection of p-adic Galois representations into nice and more manageable classes - this will be done in §13. Finally, we will see how we can use the machinery built so far to turn the study of crystalline representations, say, into the study of certain (semi-) linear algebra objects §14.

1.1. **Prerequisites:** Commutative algebra, number theory (local non-archimedean fields, ramification theory, infinite Galois theory). Strictly speaking, it will not be necessary to know algebraic geometry but some knowledge of elliptic curves would prove helpful.

1.2. **Further information:** If you want to give a talk, please send me an email at sebastian.bartling@uni-due.de in which you tell me which talk you want to give. We will meet Mondays 16:15-17:45 in N-U-4.04. Please contact me if you have questions regarding your talk and make an appointment to meet (please do not feel shy about this - reaching out for help will not affect your mark).

1.3. **References:** The main reference will be the notes of Fontaine–Ouyang [4]. They are available here: https://math.stanford.edu/~conrad/252Page/handouts/galoisrep.pdf

Another great reference are the lectures notes of Brinon–Conrad [3]. They are available here: https://math.stanford.edu/~conrad/papers/notes.pdf

Finally, for the part concerning the perfectoid theory, I really like these notes by Bhatt [2]. They are available here: https://websites.umich.edu/~bhattb/teaching/mat679w17/lectures.pdf

We will also use notes of Zink for the talk about Witt vectors. They are available here: https://www.math.uni-bielefeld.de/~zink/pDivGr1.pdf.

Here is the list of talks:

2. Introduction (Sebastian)

Introduction of the topic and distribution of the talks.

3. Structure of the Galois group of a p-adic field

Quickly recall what the p-adic numbers $\mathbb{Z}_p \subset \mathbb{Q}_p$ are. Define a *p-adic field* as a finite extension $K/\mathbb{Q}_p$. Recall the valuation ring $\mathcal{O}_K$, maximal ideal $\mathfrak{m}_K$ and residue field $\mathcal{O}_K/\mathfrak{m}_K$ (for example by summarizing [5, Chapter 1, §2, Proposition 3]). Recall that the normalized p-adic valuation on $\mathbb{Q}_p$ canonically extends to a valuation on K [5, Chapter 1, §2, Corollary 4]. Sketch why $\mathcal{O}_K$ is a complete discrete valuation ring, define the *ramification index* $e(L/K)$ of an extension L/K of p-adic fields, recall the notion of unramified, totally ramified and tamely ramified resp. wildly ramified. Discuss examples for these three different types of extensions. (a good overview of the material needed here is also given in [4, §(0.2.1)]) Introduce the absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ of K given by $\mathrm{Aut}_K(\overline{K})$ (field automorphisms of $\overline{K}$ fixing K), recall the Krull-topology on $\mathrm{Gal}(\overline{K}/K)$ [4, §0.1.2]. Explain that with this topology $\mathrm{Gal}(\overline{K}/K)$ is compact and totally disconnected [8, Tag 08ZY]. Also add some recollections on inverse limits and the inverse limit topology as in [4, §0.1.1]. Finally, discuss the short exact sequences of topological groups

$$1 \longrightarrow I_K \longrightarrow \mathrm{Gal}(\overline{K}/K) \longrightarrow \mathrm{Gal}(K^{\mathrm{ur}}/K) \longrightarrow 1$$

and

$$1 \longrightarrow I_K^{\mathrm{wild}} \longrightarrow I_K \longrightarrow I_K^{\mathrm{tame}} \longrightarrow 1.$$

Explain that $\mathrm{Gal}(K^{\mathrm{ur}/K}) \simeq \widehat{\mathbb{Z}}$ with a topological generator given Frobenius, I_K^{wild} is an inverse limit of finite groups of p-power order and why $I_K^{\mathrm{tame}} \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell(1)$ (This is the content of [4, §(0.3.2)]).

4. Basics on ℓ-adic representations

Define an ℓ-adic Galois representation of a p-adic field K (note that we allow here $\ell = p$) [4, Chapter 1, §1, Definition 1.1 and Definition 1.4]. Please explain what is the topology on $\mathrm{GL}_n(\mathbb{Q}_\ell) \simeq \mathrm{Aut}_{\mathbb{Q}_\ell}(V)$ we are using here. Give examples: cyclotomic character [4, §(1.4.(1))], Tate-module of an elliptic curve [4, §(1.4.(2))] (only sketch this example, later we will learn more about these objects - if you lack time, you can also leave it out). Show that an ℓ-adic Galois representation always admits a Galois-invariant lattice (see [4, Discussion after Chapter 1, §1, Definition 1.7]). Now assume $\ell \neq p$. Show that under this assumption the image of the wild inertia is finite (see [4, Beginning of §(1.3.1)]). State Definition (1.22) and prove Theorem (1.24) in [4].

Say what is the image of the wild inertia for p-adic cyclotomic character of $\mathbb{Q}_p$ (explain in particular why it is *not* finite).

5. $\mathbb{C}_p$

Let $K/\mathbb{Q}_p$ be a p-adic field, recall from talk 3 how to define a valuation on $\overline{K}$ extending the one given on K, explain why this valuation is *not discrete*, define the field of p-adic complex numbers as $\mathbb{C}_K := \widehat{\overline{K}}$, point out that as fields $\mathbb{C}_K = \mathbb{C}_{\mathbb{Q}_p}$ (we define $\mathbb{C}_p := \mathbb{C}_{\mathbb{Q}_p}$), state that the valuation ring $\mathcal{O}_{\mathbb{C}_K}$ is not noetherian ([8, Tag 00II]) and that the residue field is $\overline{\mathbb{F}}_p$. State that the absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ acts on $\mathbb{C}_K$. State and prove Krasner's Lemma [4, Proposition (3.3.1)], show that $\mathbb{C}_K$ is algebraically closed ([4, Corollary (3.3.2)] we only need the characteristic zero case), state and prove the Ax–Sen Lemma in characteristic zero [4, Proposition (3.3.3)], deduce that for a closed subgroup $H \subseteq G_K$, $\mathbb{C}_K^H = \widehat{L}$, where $L = \overline{K}^H$ ([4, Proposition (3.3.8)] again only characteristic zero). Next, state, without proof, the following important Theorem of Ax–Sen–Tate.

Theorem 5.1 (Ax–Sen–Tate)**.** (a) $\mathbb{C}_K^{G_K} = K$, for all $0 \neq r \in \mathbb{Z}$, $\mathbb{C}_K(r)^{G_K} = 0$,

(b) for $0 \neq r \in \mathbb{Z}$, any extension

$$0 \longrightarrow \mathbb{C}_K(r) \longrightarrow W \longrightarrow \mathbb{C}_K \longrightarrow 0$$

in the category of finite dimensional $\mathbb{C}_K$-vector spaces with continuous semi-linear G_K-action is split.

Point out that item (a) implies that there is no element $(2\pi i)_p \in \mathbb{C}_p$ such that $g((2\pi i)_p) = \chi_{\mathrm{cycl}}(g)(2\pi i)_p$ for $g \in G_{\mathbb{Q}_p}$.

6. Elliptic curves over local fields and Tate parametrization

The aim of this talk is to provide the background needed to prove the Hodge–Tate decomposition for elliptic curves with split multiplictive reduction in the next talk.

For simplicity, let L be a field of characteristic zero. Briefly introduce elliptic curves over L. For a field extension M/L, recall the points $\mathbb{P}^2(M) = (M^3 - \{(0,0,0)\})/M^*$ of projective plane (lines through the origin in M^3), an elliptic curve for us is simply the solution set $E(M) \subset \mathbb{P}^2(M)$ of the homogeneous polynomial

$$E : Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3,$$

where $a_i \in L$ and we require that $\Delta \neq 0$. You can mention that this equation can be brought into a simpler form using variable transformation. The solution set $E(M)$ has a unique point with $Z = 0$ namely $[0:1:0] \in E(M)$ and we call this the point at infinity. Recall the j-invariant $j(E)$ of E (all of this can be found in [7, Chapter III, §1]). State that $E(M)$ is abelian group with identity element $[0:1:0] \in E(M)$ and draw the picture for the group law (see [7, Chapter III, §2]).

Now assume that $L/\mathbb{Q}_p$ is a p-adic field. Explain what a minimal Weierstrass equation is [7, VII.1.Definition]. State different reduction types of elliptic curves over L as in [7, Definition VII.5] (for us *split multiplicative reduction* will be most important).

Next, we want to talk about the Tate curve where we follow [6, Chapter V, §3]. State [6, Chapter V, §3, Theorem 3.1] completely. Prove (a), you can take on faith that E_q defines an elliptic curve but explain why it has automatically split multiplicative reduction, show that the series in (c) converge as claimed in the Theorem, deduce from the complex case that the map ϕ actually takes values in $E_q(\overline{K})$; you do not have to prove the rest. Finally, state that for any p-adic field $K/\mathbb{Q}_p$, any elliptic curve E/K with split multiplicative reduction is isomorphic to the Tate curve E_q for a unique $q \in \overline{K}$ with $0 < |q| < 1$ ([6, Chapter V, §5, Theorem 5.3]).

7. Formalism of admissible representations and Hodge–Tate representations

Explain what B-representations are and what it means for them to be trivial (this is [4, §(2.1.1)], especially Definition 2.2 and Definition 2.5). Next explain the formalism of (F, G)-regular rings as in [4, §(2.2.2), Definition 2.8], give the very important [4, Definition 2.12] and state [4, Theorem 2.13(1)] and give a proof (optional).

Let $K/\mathbb{Q}_p$ be a p-adic field, then introduce $\mathbb{B}_{\mathrm{HT}}$, show it its $(\mathbb{Q}_p, G_K)$-regular, compute the invariants, give the criterion V is Hodge–Tate iff $V \otimes_{\mathbb{Q}_p} \mathbb{C}_K \simeq \bigoplus_i \mathbb{C}_K(i)$, define Hodge–Tate representations, define the Hodge–Tate numbers (see [3, Section 2, Definition 2.4.7, Definition 2.3.4]).

The main mathematical focus of this talk should be about the discussion of the following example: Let E/K be an elliptic curve with split multiplicative reduction. Then by the previous Talk 6, we know that $E \simeq E_q$ is a Tate-curve. The p-adic Tate module is $T_p(E) = \lim_p E(\overline{K})[p^n]$. Point out that it has a continuous $\mathrm{Gal}(\overline{K}/K)$-action. Use the Tate parametrization $\overline{K}^*/q^{\mathbb{Z}} \simeq E_q(\overline{K})$ to analyze the structure of the Tate module (give two basis elements and describe the Galois action on them[2]), arrive at a short exact sequence

$$0 \longrightarrow \mathbb{Z}_p(1) \longrightarrow T_p(E_q) \longrightarrow \mathbb{Z}_p \longrightarrow 0.$$

Now use the Theorem of Ax–Sen–Tate, to deduce that

$$T_p(E_q) \otimes_{\mathbb{Z}_p} \mathbb{C}_K \simeq \mathbb{C}_K(1) \oplus \mathbb{C}_K$$

as $\mathbb{C}_K$-vector spaces with a continuous semi-linear G_K-action. Deduce that $V_p(E_q) = T_p(E_q) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ is a Hodge–Tate Galois representation. If time permits and you feel comfortable, you can state the general Theorem of Faltings as on the Hodge–Tate decomposition (see [3, Theorem 2.2.3]).

8. Tilting

Define a perfectoid field [2, Definition 3.1.1], give examples (for example $\mathbb{C}_K, \widehat{\mathbb{Q}_p(\zeta_{p^\infty})}, \mathbb{F}_p(\!(t^{1/p^\infty})\!)$...see e.g. [2, Example 3.1.2]), let R be a ring (commutative with one as always), define the tilt $R^\flat$ as in [2, Definition 2.0.1], mention [2, Remark 2.0.2], mention (some of) the examples [2, Example 2.0.3] and point out that in general this operation can be quite lossfull, then cover [2, Lemma 2.0.5, Lemma 2.0.6., Remark 2.0.9, Lemma 2.0.10]. Then turn back to the case of perfectoid fields and explain [2, Lemma 3.2.2] and then cover [2, Proposition 3.2.4]. State without proof [2, Proposition 3.2.10]. Finally, state

[2]Talk to me if you get stuck on this; I did not find a reference where this calculation is done in enough details.

without proof that for a perfectoid field L and its tilt $L^\flat$, there is a canonical isomorphism of absolute Galois groups

$$\mathrm{Gal}((L^\flat)^{\mathrm{sep}}/L^\flat) \simeq \mathrm{Gal}(L^{\mathrm{sep}}/L)$$

(see [2, Theorem 3.2.8]).

9. Witt vectors

Define the Witt-polynomials $W_n(X_0,...,X_n) = X_0^{p^n} + pX_1^{p^{n-1}} ... + p^n X_n$, state Lemma 2.1 [9], explain the comment after that Lemma in [9], then prove [9, Proposition 2.2], and state [9, Corollary 2.3], then give [9, Theorem 2.4, Definition 2.5]. Explain the Verschiebung [9, Lemma 2.6] and please give the proof since this is a general proof strategy. Then only state [9, Proposition 2.7, Lemma 2.8]. Introduce the Frobenius homomorphism [9, Definition 2.9, Lemma 2.10]. Then state [9, Proposition 2.13] (a proof is optional). Finally turn to [9, Proposition 2.15]; this is the main result of this talk. Please explain as much as possible of the proof given what you have covered before. As a final remark, prove that $W(\mathbb{F}_p) \simeq \mathbb{Z}_p$ and that $W(\mathbb{F}_q)[1/p]$ is an unramified extension of $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$. More generally, show that for a perfect field k, $W(k)$ is a complete discrete valuation ring with uniformizer p and residue field k.

10. $\mathbb{A}_{\mathrm{inf}}$

Define $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L) = W(\mathcal{O}_L^\flat)$ for a perfectoid field L, show that it is a non-noetherian local domain (Namely, we have $\mathcal{O}_L^\flat \subset L^\flat$ so that $W(\mathcal{O}_L^\flat) \subset W(L^\flat)$ but $L^\flat$ is a perfect field by Talk 8 and by Talk 9 we know that the Witt vectors of such a field are a complete discrete valuation ring. It is local because $\ker(\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L^\flat) \to \mathcal{O}_L^\flat \to \mathcal{O}_L^\flat/\mathfrak{m}_L^\flat)$ is the unique maximal ideal; if an element is not in this set, then it is invertible modulo p (recall that $\mathcal{O}_L^\flat$ is local) so since $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L^\flat)$ is p-adic, it is invertible itself. It is non-noetherian because we learnt that $\mathcal{O}_L^\flat$ is non-noetherian and quotients of noetherian rings would be noetherian), explain that $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$ has a lift of Frobenius $\varphi\colon \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L) \to \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$ given by $W(\mathrm{Frob}_{\mathcal{O}_L^\flat})$ that is an isomorphism, construct Fontaine's map $\theta\colon \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L) \to \mathcal{O}_L$ [3, Beginning of Section 4.4, Lemma 4.4.1], show that it is surjective, show that the kernel is principal [3, Proposition 4.4.3], describe $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\mathbb{Q}_p(\zeta_{p^\infty})})$, state and prove the universal property (optional).

11. $\mathbb{A}_{\mathrm{cris}}$ and $\mathbb{B}_{\mathrm{cris}}$

Let $L/\mathbb{Q}_p$ be a perfectoid field, consider $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$ from the previous talk and let $\xi \in \ker(\theta)$ be a generator, define

$$\mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L) = \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{\xi^n}{n!}] \subseteq \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}]$$

as the $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$-subalgebra of $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}]$ generated by the so-called *divided powers* $\xi^n/n!$ (for now we have to take these on faith; there is however a conceptual explanation for considering these elements). Point out that this ring is independent of the choice of a generator of $\ker(\theta)$ and that one has to say somethhing for why this is indeed a ring. Sketch roughly why

$$\mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L) \simeq \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[T_0, T_1, ...]/(pT_0 - \xi^p, pT_1 - T_0^p, ...)$$

(see for example [1, Lemma 8.8]) State without proof that $\mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L)$ is p-adically separated. Then $\mathbb{A}_{\mathrm{cris}}(\mathcal{O}_L)$ is defined to be the p-adic completion of $\mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L)$. We obtain $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L) \subseteq \mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L) \subseteq \mathbb{A}_{\mathrm{cris}}(\mathcal{O}_L)$. $\mathbb{A}_{\mathrm{cris}}(\mathcal{O}_L)$ is a p-torsionfree p-adic ring which has a universal property (optional). Show that φ on $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}]$ stablizes $\mathbb{A}_{\mathrm{cris}}{}^{\circ}(\mathcal{O}_L)$ [3, Lemma 9.1.7] and therefore extends to a Frobenius lift φ on $\mathbb{A}_{\mathrm{cris}}(\mathcal{O}_L)$. State without proof that this map is *injective*. Let $K/\mathbb{Q}_p$ be a p-adic field. Now take $L = \mathbb{C}_K$. State that $\mathbb{A}_{\mathrm{cris}}(\mathcal{O}_{\mathbb{C}_K})$ admits a continuous action by $\mathrm{Gal}(\overline{K}/K)$ [3, Proposition 9.1.2]. Consider a compatible system of p^n-th root of unity $\zeta_{p^n} \in \mathcal{O}_{\mathbb{C}_K}$, this gives $\varepsilon = (\zeta_{p^n})_n \in \mathcal{O}_{\mathbb{C}_K}^\flat$. We can then consider $[\varepsilon] \in \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\mathbb{C}_K})$. Consider the formal sum $\log([\varepsilon]) = \sum_{n\geq 1}(-1)^{n+1}\frac{([\varepsilon]-1)^n}{n}$. Show that $t \in \mathbb{A}_{\mathrm{cris}}(\mathcal{O}_{\mathbb{C}_K}) \subseteq \mathbb{A}_{\mathrm{cris}}(\mathcal{O}_{\mathbb{C}_K})[\frac{1}{p}] =: \mathbb{B}_{\mathrm{cris}}{}^{+}(\mathcal{O}_{\mathbb{C}_K})$ [3, Proposition 9.1.3], define $\mathbb{B}_{\mathrm{cris}}(\mathcal{O}_{\mathbb{C}_K}) = \mathbb{B}_{\mathrm{cris}}{}^{+}(\mathcal{O}_{\mathbb{C}_K})[\frac{1}{t}]$. Explain that $\varphi(t) = pt$ and that for all $g \in \mathrm{Gal}(\overline{K}/K)$, $g(t) = \chi_{\mathrm{cycl}}(g) \cdot t$. For this, see [3, Section 9.1].

12. $\mathbb{B}_{\mathrm{dR}}$ and $\mathbb{B}_{\mathrm{st}}$

Let $L/\mathbb{Q}_p$ be a perfectoid field. Recall $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$, define $\mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_L) = \widehat{\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}]}_{\ker(\theta)}$ ($\ker(\theta)$-adic completion of $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}]$). We have that $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)[\frac{1}{p}] \subset \mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_L)$. State that there exists an embedding

$$\mathbb{B}_{\mathrm{cris}}{}^+(\mathcal{O}_L) \to \mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_L).$$

State that we may then (non-uniquely) write $\alpha \in \mathbb{A}_{\mathrm{cris}}(\mathcal{O}_L)$ as $\alpha = \sum_{n\geq 0} a_n \frac{\xi^n}{n!} \in \mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_L)$ where $a_n \in \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_L)$ and $a_n \to 0$ for $n \to \infty$ p-adically. Define $\mathbb{B}_{\mathrm{dR}}(\mathcal{O}_L) = \mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_L)[\frac{1}{\ker(\theta)}]$. Show that this is a discrete valuation field. Now let $L = \mathbb{C}_K$ and let $t = \log([\varepsilon]) = \sum_{n\geq 1}(-1)^{n+1}\frac{([\varepsilon]-1)^n}{n}$. Show that $t \in \mathbb{B}_{\mathrm{dR}}{}^+(\mathcal{O}_{\mathbb{C}_K})$ and that it is a uniformizer. That is Fontaine's p-adic $2\pi i$. Finally, introduce $\mathbb{B}_{\mathrm{st}}$. A reference for this is [3, Section 4.4].

13. Crystalline, semistable and de Rham representations

State that $\mathbb{B}_{\mathrm{cris}}$, $\mathbb{B}_{\mathrm{st}}$ and $\mathbb{B}_{\mathrm{dR}}$ are G_K-regular, define crystalline, semistable and de Rham representations, discuss one-dimensional crystalline representations for $K = \mathbb{Q}_p$, discuss in detail why the Tate-module of the Tate-elliptic curve is a semistable Galois representation and describe $D_{st}(V_p(E_q))$. Mention that elliptic curves with good reduction give examples of crystalline Galois representations (however we will not be able to prove it). A reference for this is [3, Section 8 and Section 9].

14. weakly-admissible filtered Isocrystals and weakly-admissible implies admissible

Define isocrystals, define weakly-admissible isocrystals, state Theorem B in [4], namely that weakly-admissible is the same as admissible. Also define filtered (φ, N)-modules. We will not be able to prove this. State the fundamental exact sequence of p-adic Hodge theory. Show the degree one case (i.e. the one concerning $\varphi = p$). Show that admissible implies weakly-admissible. A reference for this is [3, Section 8 and Section 9].

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